

Practice problems on pointers, functions and arrays

Problem 1

Sum of Array Elements

Write a function to calculate the sum of elements in an array using pointers.

```
int arraySum(int *arr, int size) {  
    // Your code here  
}
```

Input: An integer array {1, 2, 3, 4, 5} and its size

Output: The sum of the elements (e.g., 15)

Problem 2

Reverse an Array

Write a function to reverse the elements of an array using pointers.

```
c  
Copy code  
void reverseArray(int *arr, int size) {  
    // Your code here  
}
```

Input: An array {1, 2, 3, 4, 5}

Output: The reversed array {5, 4, 3, 2, 1}

Problem 3

Write a function to count the number of vowels in a string using pointers.

```
int countVowels(char *str) {  
    // Your code here  
}
```

Input: "Pointers"

Output: 3

Problem 4

Remove Duplicates from an Array

Write a function that removes duplicate elements from a sorted array using pointers.

```
int removeDuplicates(int *arr, int size) {  
    // Your code here  
}
```

Input: {1, 1, 2, 3, 3, 4}

Output: {1, 2, 3, 4}

Problem 5

Write a function to check if a given string is a palindrome using pointers.

```
c  
Copy code  
int isPalindrome(char *str) {  
    // Your code here  
}
```

Input: "madam" or "hello"

Output: 1 for a palindrome, 0 otherwise